7. Delete by value

8. Exit

Enter choice: 7

Enter value to delete: 30

1. Display
2. Insert at front
3. Insert at end
4. Insert at position
5. Delete at front
6. Delete at end
7. Delete by value
8. Exit

Enter choice: 1

List: 10 ⟷ 45 ⟷ 20 ⟶ NULL

1. Display
2. Insert at front
3. Insert at end
4. Insert at position
5. Delete at front
6. Delete at end
7. Delete by value
8. Exit

Enter choice: 8

08.12.2025

Write a program a) to Construct a binary search tree.
b) to traverse the tree using all the methods i.e in order, pre-order and postorder.
c) to display the elements.

```
#include<stdio.h>
#include<stdlib.h>
struct Node {
    int data;
    struct Node *left, *right;
};

struct Node* CreateNode(int value){
    struct Node *newNode = (struct Node*) malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->left = newNode->right = NULL;
    return newNode; }
```

```
struct Node* insert(struct Node *root, int value){
    if(root==NULL)
        return createNode(value);
    if (value<root->data)
        root->left=insert(root->left,value);
    else if (value>root->data)
        root->right= insert(root->right,value);
    return root;
}

void inorder (struct Node *root) {
    if(root==NULL)
        return;
    inorder(root->left);
    printf("%d", root->data);
    inorder(root->right);
}
void preorder ( struct Node *root ) {
    if(root==NULL)
        return;
    printf("%d",root->data);
    preorder(root->left);
    preorder(root->right);
}
void postorder (struct Node *root) {
    if(root==NULL)
        return;
    postorder(root->left);
    postorder(root->right);
    printf("%d", root->data);
}
void display (struct Node *root) {
    printf(" BST elements (Inorder) : ");
    inorder(root);
    printf("\n");
}
int main() {
    struct Node * root= NULL;
    int choice, value;
while (1) {
    printf(" \n-- Binary search Tree Menu --\n" );
```

```
printf("1. Insert into BST\n");
printf("2. Inorder Transversal \n");
printf("3. pre-order Traversal \n");
printf ("4. postorder Traversal \n");
printf ("5. Display BST \n");
printf("6. Exit \n");
printf("Enter choice: ");
scanf ("%d", &choice);
switch (choice) {
        case 1 :
            printf("Enter value to insert :");
            scanf ("%d", &value);
            root = insert(root, value);
            break;
        case 2 :
            printf (" Inorder Traversal : ");
            inorder (root);
            printf("\n");
            break;
        case 3 :
            printf (" preorder Traversal :");
            preorder (root);
            printf ("\n");
            break;
        case 4 :
            printf (" postorder Traversal : ");
            postorder (root);
            printf ("\n");
            break;
        case 5 :
            display (root);
            break;
        case 6 :
            exit(0);
        default :
            printf(" Invalid choice ! Try again\n");
    }
}
return 0;
}
```

Output:

---Binary Search Tree Menu---

1. Insert into BST

2. Inorder Traversal

3. Preorder Traversal

4. Postorder Traversal

5. Display BST

6. Exit

Enter choice :1

Enter value to insert : 40.

Enter choice :1

Enter value to insert :25

Enter choice :1

Enter value to insert :60

Enter choice:2

Inorder Traversal: 25 40 60

Enter choice :3

Preorder Traversal : 40 25 60

Enter choice :4

Postorder Traversal: 25 60 40

Enter choice :5

BST element (Inorder): 25 40 60

Enter choice 6.